

♦ P3: Password Cracking using SHA-256

Aim

To understand password hashing and cracking using SHA-256.

Tools Required

Kali Linux (WSL), John the Ripper, Wordlist (for a dictionary-based attack)

Procedure

Step 1: Install John the Ripper

- Install John tool in Kali Linux.

```
(kali@HP15s)-[~]  
$ sudo apt install john
```

Step 2: Generate SHA-256 Hash

- Generate a hash of a sample password.
- Save the hash into a file.

```
(kali@HP15s)-[~]  
$ nano hash.txt
```

Step 3: Prepare Wordlist

- Ensure wordlist is available in system.

```
sudo apt install wordlists  
sudo gzip -d /usr/share/wordlists/rockyou.txt.gz
```

Step 4: Crack the Hash

- Run John the Ripper with SHA-256 format.

- Perform dictionary attack using wordlist.

```
(kali@HP15s)-[~]  
$ john --format=raw-sha256 --wordlist=/usr/share/wordlists/rockyou.txt hash.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (Raw-SHA256 [SHA256 256/256 AVX2 8x])  
Warning: poor OpenMP scalability for this hash type, consider --fork=12  
Will run 12 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
password123 (?)  
lg 0:00:00:00 DONE (2026-03-29 09:47) 14.28g/s 2808Kp/s 2808Kc/s 2808KC/s 123456..piggy9  
Use the "--show --format=Raw-SHA256" options to display all of the cracked passwords reliably  
Session completed.
```

Step 5: View Cracked Password

- Display the cracked password.

```
(kali@HP15s)-[~]  
$ john --show --format=Raw-SHA256 hash.txt  
?:password123  
  
1 password hash cracked, 0 left
```

Result

Successfully cracked a SHA-256 hashed password using John the Ripper.